

6.2.2 8-bit data

8-bit data is partially user defined. Each 8-bit unit is transmitted with the *b1* “first”, up to *b8* “last”. (NOTE: at a lower protocol level: in 16-QAM 4 bits are sent in parallel, 64-QAM: 6 bits in parallel, 256-QAM: 8 bits in parallel, 1024-QAM: 10 bits in parallel.)

- Pad character:
 - * Character data: CR (0x0D in 8 bits (octet), after bits reversal) or 0x00.
 - * Non-character data: the pad octet is user defined and NLIDs are ignored.
- Character table (if characters; support is user defined):
 - ISO/IEC 10646 [10] as UTF-8, 1 to 4 octets per character, if NLIDs are both 0x00 (default). If the National Language ID are both the same value and 0x10 or greater: the 8-bit version of the 7-bit alphabet (section A.4) having that ID. If the NLIDs are in 0x01..0x0F, the character table is user defined. NLID is available for SMS only; for other protocols the character table is user defined.

The UTF-8 text should be in NFC (Normal Form C) (Unicode). For UTF-8, the mappings in subclause 6.2.3 apply when sending as well as receiving an SMS message.

For the 8-bit version of a 7-bit alphabet: for 0x00 use 0x1B, for 0x1B00..0x1B7F (SS2 table, two 7-bit code units) use 0x80..0xFF (one 8-bit code unit with 1 as most significant bit) and for 0x1B1B00..0x1B1B7F (SS3 table, three 7-bit code units) use 0x9B00..0x9B7F (two 8-bit code units). (The 7-bit SS3, 0x1B1B (two 7-bit code units, 14 bits), is in the 8-bit version 0x9B, 8 bits.) Note: the bit reversal for 8-bit data is always on octets.

When splitting a message text into submessages, there must be no cut inside of a UTF-8 byte sequence. For 8-bit versions of alphabets with National Language ID 0x10 or larger: there must be no cut between 0x9B and the follow-on byte. When not used for characters or the character table is user defined, the split rule is user defined.